

From Transfer Learning to Deep Transfer Learning, Taxonomy

It is possible to categorize Deep Transfer Learnings (DTLs) in different ways by various criteria, similar to Transfer Learnings. DTLs can be divided into two categories of homogeneous and heterogeneous based on the homogeneity of source and target data. However, this categorization can be conducted differently because it is subjective and relative. For example, a dataset of X-ray photos can be considered heterogeneous to a dataset of tree species photos when the comparison domain is limited to only image data. In contrast, it can be considered homogeneous to the same tree species photo dataset when the domain consists of audio and text datasets.

Moreover, DTLs can be categorized into three groups based on label-setting aspects: (i) transductive, (ii) inductive, and (iii) unsupervised. Briefly, transductive is when only the source data is labeled; if both source and target data are labeled it is inductive; if none of the data are labeled, it is unsupervised deep transfer learning.

Refs. mention and define another categorization of DTLs through the aspect of applied approaches. They similarly categorized DTLs into four groups of: (i) instance-based, (ii) feature-based/mapping-based, (iii) parameter-based/model-based, and (iv) relational-based/adversarial-based approaches. Instance-based transfer learning approaches are based on using the selected parts of instances (or all) in source data and applying different weighting strategies to be used with target data. Feature-based approaches map instances (or some features) from both source and target data into more homogeneous data. Further, the survey divides the feature-based category into asymmetric and symmetric feature-based transfer learning subcategories. “Asymmetric approaches transform the source features to match the target ones. In contrast, symmetric approaches attempt to find a common latent feature space and then transform both the source and the target features into a new feature representation”. The model-based (parameter-based) methods are about using the obtained knowledge in the model (network) with different combinations of pre-trained layers: freezing some and/or finetuning some and/or adding some fresh layers. Relational/adversarial-based approaches focus on extracting transferable features from both source and target data either using the logical relationship or rules learned in the source domain or by applying methods inspired by generative adversarial networks (GAN). **Figure 1** shows the taxonomy of the above-mentioned categories.

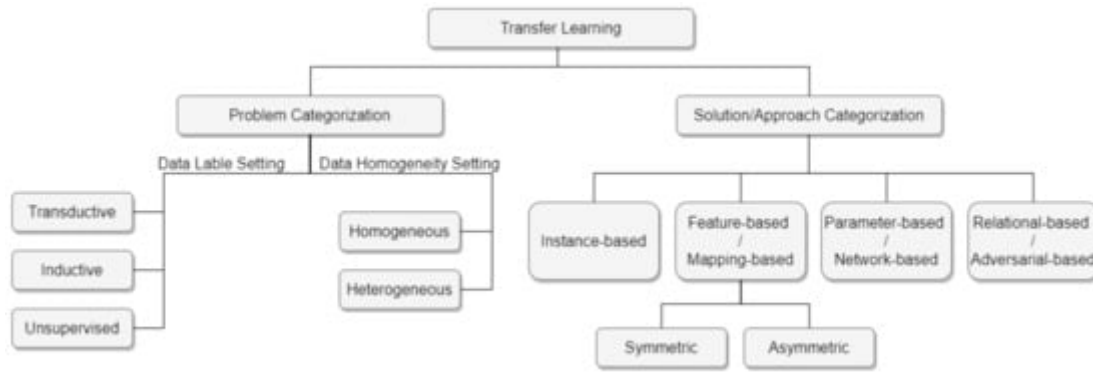


Figure 1. Taxonomy of Transfer Learning which is extendable to Deep Transfer Learning as well.

Other than the model-based and adversarial-based approaches, all other categories have been explored deeply during the last couple of decades for different ML techniques known as domain adaptation or transfer learning. However, most of those techniques are still applicable to deep transfer learning (DTL) as well. Model-based (parameter-based) approaches are the most applied techniques in DTL since they can tackle the domain adaptation between the source and target data by adjusting the network (model). In other words, deep transfer learning is mainly focused on model-based approaches. Remarkably, model-based approaches in deep learning models can even tackle the adaptation of a very distant source and target data.

In deep transfer learning (DTL), different techniques are applied for model-based approaches, although generally, they are combinations of pre-training, freezing, finetuning, and/or adding a fresh layer(s). A deep learning network (DL model) trained on source data is called a pre-trained model consisting of pre-trained layers. Freezing and finetuning are techniques using some or all layers of pre-trained models to train the model on target data. Freezing some layers means the parameters/weights will not change and are constant values for frozen layers from a pre-trained model. Finetuning means the parameters/weights are initialized with the pre-trained values instead of random initialization for the whole network or some selected layers. Another recent DTL technique is based on freezing a pre-trained model and adding new layers to that model for training on target data; Google's deep mind project introduces this technique in 2016 as Progressive Learning/progressive neural networks (PNNs).

The concept of progressive learning mimics human skill learning, which is adding a new skill on top of previously learned skills as a foundation to learn a new one. E.g., a child learns how to run after learning to crawl and walk and using all the skills obtained in the process. Similarly, PNNs prevent catastrophic forgetting in DTL versus finetuning techniques by freezing the

whole pre-trained model and learning (adjusting to) the new task by training the newly added layers on top of the previously trained layers.

In deep learning models, usually, the earlier layers conduct the feature extraction at a high level of detail, further layers towards the end extract the information and conceptualize the given data, and lateral layers conduct the classifications or predictions. For instance, in the image-related model, the earlier layers of CNN extract the edges, corners, and tiny patches of a given image. Further layers put those details together to detect objects or faces, and the lateral layers, usually fully connected layers, conduct the classification. Given this process, the most effective and efficient approach for DTL, to our knowledge, is to freeze the earlier and middle layers from a related pre-trained model and finetune the lateral layers for the new task/dataset. Similarly, the new layers are added to the last part of a pre-trained model in progressive learning.

Nonetheless, some other research in this area use combinational and sophisticated methods to tackle transfer learning in deep learning such as ensembled networks, weighting strategies, etc.. However, to our knowledge, the search for recent advancements in DTL for practical tasks ends up with methods based on mostly the model-based and limited number of adversarial-based approaches.